CMPT 280

Topic 17: B+ Trees and B Trees

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References

- Textbook, Chapter 17

Exercise 1

- B+ trees: generalization of 2-3 trees.
- Insertion and deletion algorithms are the same, except more children are allowed.
- B+ tree of order b — internal nodes have between $\lceil b/2 \rceil$ and b children (root may have between 2 and b).
- Leaf nodes are collapsed — each can hold between $\lceil b/2 \rceil - 1$ and $b - 1$ keyed data items.
- 2-3 Tree is (more or less) a B+ tree of order 3, but not identical due to the way we defined 2-3 trees (see Section 17.1.1 in the textbook).
- Build a B+ tree of order 5 using the keys:
  20, 2, 7, 4, 14, 8, 12, 9, 17, 1, 18, 15, 3, 19,
  11, 10, 13, 6, 16

Exercise 2

- B-trees: generalizations of ordered binary trees.
- Both internal and leaf nodes contain data items.
- Insertion/deletion algorithms similar to B+ trees - split, give/steal.
- Internal nodes in B-trees of order b have nodes have between $\lceil b/2 \rceil$ and b children (root may have between 2 and b)
- Nodes contain between $\lceil b/2 \rceil - 1$ and $b - 1$ data items (root may have between 1 and $b - 1$ data items).
- Repeat exercise 1, but build a B-tree of order 6.

Next Class

- Next class reading: Chapter 18: k-D trees.